Physics-Based Lithium-Ion Battery Pack Model Development

Abstract

A Python-based computational framework for physics-based modeling for lithium-ion battery packs is developed to capture cell-to-cell heterogeneity, thermal gradients, and their coupled effects on cell aging and current distribution. The modeling approach extends single-cell electrochemical models (SPMe) to arbitrary series-parallel pack topologies. The framework explicitly solves for Solid Electrolyte Interphase (SEI) layer growth, coupling solvent diffusion and side reaction kinetics with thermal states. The model utilizes a Finite Difference Method (FDM) for solid-phase diffusion and a Finite Volume Method (FVM) with harmonic mean properties for electrolyte transport, ensuring robust handling of material discontinuities. A grid-based thermal interconnection matrix calculates conductive heat transfer between adjacent series and parallel neighbors. The model is validated against the open-source PyBaMM library at the cell level, ensuring electrochemical accuracy. While the current implementation utilizes a nested optimization approach (`fsolve`) to handle Kirchhoff's laws, highlighting a computational bottleneck, it successfully captures the nonlinear divergence of cell states due to thermal gradients. Results indicate that cells subject to higher temperatures experience accelerated SEI growth, leading to resistance rise and current redistribution. This work establishes the foundational physics and interconnection logic required for a high-fidelity pack model. Future development targets a computationally efficient, high-fidelity Python framework designed for plug-and-play integration of new physics.

Keywords: Physics-based battery modeling, Series-Parallel Topology, Thermal Interconnection Matrix, SEI degradation, Current distribution, FDM/FVM Discretization

1 Introduction

1.1 Problem Statement and Motivation

Lithium-ion batteries are critical energy storage devices for electric vehicles (EVs) and grid-scale applications. However, a significant disconnect exists between the high fidelity of single-cell physics-based models and the simplified models currently used for pack-level management [1-3]. While the Doyle-Fuller-Newman (DFN) model and its simplified variants (SPM/SPMe) accurately describe internal cell states—such as lithium concentration gradients and local potentials—scaling these physics to a full battery pack remains a formidable computational challenge [4].

Current pack-level modeling is dominated by Equivalent Circuit Models (ECMs) due to their low computational cost. However, ECMs are phenomenological; they cannot predict internal physical states or degradation mechanisms like lithium plating and Solid Electrolyte Interphase (SEI) growth based on first principles. Consequently, they fail to capture the complex, coupled dynamics of cell heterogeneity, where manufacturing variations and thermal gradients cause uneven current distribution and accelerated localized aging.

Furthermore, existing attempts to upscale physics-based models often suffer from rigid topological assumptions. Most literature focuses solely on parallel-connected cells to study current imbalance, neglecting the voltage imbalances and thermal propagation inherent in series-connected strings. There is a distinct lack of flexible, open modeling frameworks that allow for arbitrary series-parallel configurations while permitting the modular addition of new physics (such as degradation equations) without restructuring the entire solver. This work addresses these gaps by developing a modular, control-oriented physics-based pack framework that captures electrochemical-thermal-degradation coupling across complex pack topologies.

1.2 Literature Review

1.2.1 The Limitations of Equivalent Circuit Models (ECMs)

The industry standard for battery management systems (BMS) remains the Equivalent Circuit Model, primarily due to its simplicity and ease of parameter identification [1, 5]. Authors such as Plett [1] and Hu et al. [19] have established robust frameworks for state-of-charge (SOC) and state-of-health (SOH) estimation using ECMs. However, these models treat the battery as a "black box," relying on look-up tables and empirical resistors and capacitors. As noted by varying studies, ECMs struggle to predict performance outside their calibration range and fail to provide insight into internal limiting mechanisms, such as electrolyte depletion or electrode saturation, which are critical for fast-charging protocols.

1.2.2 Upscaling Physics-Based Models: The Computational Bottleneck

To overcome ECM limitations, researchers have attempted to upscale the DFN model. However, a full DFN simulation for a standard EV pack (e.g., 96 series $\times$ 70 parallel cells) involves solving thousands of coupled partial differential equations (PDEs), which is computationally prohibitive for real-time applications [20].

To mitigate this, Model Order Reduction (MOR) techniques are frequently employed. The Single Particle Model (SPM) and its electrolyte-enhanced version (SPMe), derived by Marquis et al. [9-10], offer a compromise, reducing the spatial states while retaining electrochemical accuracy. However, typical implementations of SPMe are restricted to single-cell analysis. When applied to packs, standard approaches often assume a "representative cell," implying all cells in the pack behave identically. This assumption collapses when thermal gradients exist, as Arrhenius-dependent parameters cause significant divergence in cell behavior [6, 8].

1.2.3 Gaps in Topology and Heterogeneity: Parallel vs. Series

A critical review of recent literature reveals a bias toward specific electrical topologies.

- **Parallel-Only Studies:** Significant attention has been paid to parallel-connected cells [17-18, 22]. Research by Drummond et al. [17] and Bruen et al. focused on solving Kirchhoff's laws for parallel strings to quantify current imbalances caused by impedance mismatch. While valuable, these studies typically neglect the series connection dynamics found in high-voltage packs.

- **Series-Only Studies:** Conversely, works by Weaver et al. [5, 15] have explored series strings, highlighting voltage imbalance and thermal gradients. However, these often simplify the physics to 0D models or neglect the complex current redistribution that occurs when series modules are placed in parallel (SP topology).

There is a scarcity of unified frameworks capable of simulating arbitrary Series-Parallel (SP) or Parallel-Series (PS) configurations with full electrochemical coupling.

1.2.4 The Lack of Modular Frameworks for Degradation

Finally, current literature lacks flexible frameworks that facilitate the addition of coupled degradation physics at the pack level. Most existing physics-based pack models are hard-coded for specific chemistries or geometries. Integrating new degradation mechanisms—such as the coupled mechanical-electrochemical fatigue or temperature-dependent lithium plating—often requires rewriting the core solver [12-14]. Existing commercial tools allow for pack modeling but are often closed-source, preventing the integration of novel user-defined PDEs for aging. This work proposes a modular state-space structure where electrochemical (SPMe), thermal, and degradation (SEI) sub-models are coupled via a scalable interconnection matrix, addressing the need for a flexible, high-fidelity research platform.

1.3 Contributions and Paper Organization

The primary contributions of this work are:

1. **Integrated Pack-Level Electrochemical-Thermal-Degradation Model**: Extension of SPMe-based cell models to the pack level with explicit treatment of electrical interconnection (Kirchhoff's laws) and coupled SEI growth dynamics.

2. **Validation against PyBaMM**: Verification of the core cell-level electrochemical physics against the industry-standard PyBaMM library to ensure baseline accuracy before pack-level upscaling.

3. **Nonlinear Current Distribution Algorithm**: Implementation of `fsolve`-based optimization to solve nonlinear Kirchhoff's law equations for current distribution, accounting for temperature-dependent resistance and nonlinear open-circuit potential.

4. **Grid-Based Thermal Interconnection**: Implementation of a conductive thermal matrix that models heat transfer between adjacent cells in both series and parallel directions, enabling the study of hotspot propagation.

5. **Validation Against Coupled Phenomena**: Model results demonstrating that pack-level aging, voltage response, and energy capacity differ significantly from single-cell predictions due to thermal retroactivity and current imbalance.

The remainder of this paper is organized as follows. Section 2 develops the governing equations for electrochemical, thermal, and aging dynamics at the cell level. Section 3 details the discretization strategy and matrix implementation. Section 4 describes the pack-level coupling framework. Section 5 discusses the nonlinear current distribution algorithm. Section 6 provides simulation results and validation. Section 7 discusses computational efficiency. Finally, Section 8 outlines the limitations and future roadmap for high-fidelity code development.

2 Cell-Level Governing Equations

2.1 Electrochemical Dynamics

Each lithium-ion cell is modeled using the single particle model with electrolyte effects (SPMe) framework. The electrochemical dynamics consist of three coupled partial differential equations governing lithium concentration in solid particles (anode and cathode) and lithium-ion concentration in the electrolyte.

2.1.1 Solid-Phase Lithium Diffusion

The solid-phase lithium concentration in electrode particles is governed by Fick's diffusion law:

$$\frac{\partial c_{s,j}}{\partial t} = \frac{D_{s,j}}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial c_{s,j}}{\partial r}\right), \quad j \in \{n, p\} \tag{1}$$

with boundary conditions:

$$\left.\frac{\partial c_{s,j}}{\partial r}\right|_{r=0} = 0, \quad -D_{s,j}\left.\frac{\partial c_{s,j}}{\partial r}\right|_{r=R_{s,j}} = \frac{I_{app}}{F \cdot a_{s,j} \cdot L_j} \tag{2}$$

where $c_{s,j}$ is the lithium concentration in solid phase [mol/m^3], $D_{s,j}$ is the solid diffusion coefficient [m^2/s], r is the radial coordinate in the particle, $R_{s,j}$ is the particle radius, I_{app} is the applied current density [A/m^2], F is Faraday's constant, $a_{s,j}$ is the specific surface area [m^2/m^3], and L_j is the electrode thickness.

2.1.2 Electrolyte-Phase Ion Transport

The electrolyte ionic concentration and potential are governed by coupled equations. The lithium-ion concentration in the electrolyte follows:

$$\epsilon_e \frac{\partial c_e}{\partial t} = \frac{\partial}{\partial x}\left(D_e^{eff} \frac{\partial c_e}{\partial x}\right) + \left(1 - t_0^+\right) \frac{g_{e,j} I_{app}}{A L_j F} \tag{3}$$

where ϵ_e is the electrolyte volume fraction, c_e is the lithium-ion concentration in electrolyte, D_e^{eff} is the effective electrolyte diffusion coefficient, t_0^+ is the transference number, and $g_{e,j}$ is a geometric factor (1 for anode, -1 for cathode, 0 for separator).

2.1.3 Cell Voltage

The cell voltage is calculated from the difference in electrode potentials minus the ohmic drop across the electrolyte and SEI film:

$$V_{cell}^{(k)} = U_p(c_{s,p}^{surf}) - U_n(c_{s,n}^{surf}) - \eta_p - \eta_n - \Delta\Phi_e - \Delta\Phi_s - I_{app} R_{sei} \tag{4}$$

where U_j is the open-circuit potential of electrode j, η_j is the overpotential (calculated from Butler-Volmer equation), R_{sei} is the SEI layer resistance, and R_{el} is the electrolyte resistance.

The overpotential is approximated from the Butler-Volmer equation:

$$\eta_j = \frac{2RT}{F} \sinh^{-1}\left(\frac{I_{app}}{2 a_{s,j} L_j j_{0,j}} \right) \tag{5}$$

where $j_{0,j}$ is the exchange current density (function of surface lithium concentrations and temperature), R is the gas constant, and T is temperature.

2.2 Thermal Dynamics

Heat generation in a cell arises from both reversible (entropic) effects and irreversible (Joule and overpotential) effects. The temperature evolution of cell k, denoted as $T_N^{(k)}$, is governed by the balance of heat generation, conduction between cells, and convection to the ambient environment:

$$\frac{dT_N^{(k)}}{dt} = \frac{Q_{gen}^{(k)} + Q_{cond}^{(k)} - Q_{conv}^{(k)}}{C_{th}} \tag{6}$$

where $Q_{gen}^{(k)} = I^{(k)}(V_{oc}^{(k)} - V_{cell}^{(k)})$ represents the irreversible heat generation derived from the difference between the open circuit voltage and the terminal voltage. This ensures the model captures the full electrical loss converted to heat.

2.3 Degradation Dynamics: SEI Layer Growth

Solid electrolyte interphase (SEI) layer growth is the dominant degradation mechanism. The framework explicitly solves for SEI thickness L_{sei} and solvent concentration c_{solv}.

2.3.1 Solvent Concentration in SEI

Solvent diffusion through the SEI layer is modeled on a normalized domain $\xi \in [0, 1]$ accounting for the moving boundary of the growing SEI thickness:

$$\frac{\partial c_{solv}}{\partial t} = \frac{D_{solv}}{L_{sei}^2} \frac{\partial^2 c_{solv}}{\partial \xi^2} + \frac{\xi}{L_{sei}} \frac{dL_{sei}}{dt} \frac{\partial c_{solv}}{\partial \xi} \tag{7}$$

2.3.2 SEI Layer Growth Rate

The SEI growth is driven by the side reaction current density i_s:

$$\frac{dL_{sei}}{dt} = \frac{i_s M_{sei}}{2F\rho_{sei}} \tag{8}$$

2.3.3 Side Reaction Current

The side reaction current i_s is modeled using a specific coupled kinetic-diffusion expression implemented in the code that captures the dependence on SEI thickness and surface overpotential:

$$i_s = A_{kin} \exp(-K_{tun} L_{sei}) \exp\left(-\frac{0.55F}{RT}(\phi_{s,n} - U_s)\right) + A_{diff} \frac{1}{L_{sei}} \exp\left(-\frac{U_n F}{RT}\right) \tag{9}$$

where $A_{kin} = 165.96 \times 10^{-6}$ A/m^2 is the kinetic prefactor, $K_{tun} = 6.3 \times 10^9$ m^{-1} is the tunneling factor, and A_{diff} represents the diffusion-limited contribution scaled by the Faraday constant and diffusion coefficients.

2.3.4 Capacity Fade

The loss in cell capacity due to SEI layer growth is tracked through lithium consumption:

$$\frac{dQ_{loss}}{dt} = i_s \cdot a_{s,n} \cdot L_n \tag{10}$$

Integrating this over time gives the cumulative capacity loss as a percentage of nominal capacity.

3 Discretization Strategy and Implementation

3.1 Finite Difference Method for Solid-Phase Diffusion

The solid-phase lithium diffusion equation is discretized using the **Finite Difference Method (FDM)** on a nonuniform grid. FDM is chosen for its simplicity and direct mapping to the spherical diffusion operator in the particle domain.

3.1.1 Discretization Scheme

The second spatial derivative in spherical coordinates is discretized using a second-order central difference scheme. For a node i (where i ranges from 1 to N_r), the discretized equation becomes:

$$\frac{\partial c_{s,j,i}}{\partial t} = \frac{D_{s,j}}{\Delta r^2} \left[\left(1 + \frac{1}{i}\right) c_{s,j,i+1} - 2c_{s,j,i} + \left(1 - \frac{1}{i}\right) c_{s,j,i-1} \right] \tag{11}$$

This formulation naturally handles the singular term at $r = 0$ through symmetry boundary conditions ($\partial c / \partial r = 0$) and incorporates the surface flux at $r = R_s$.

3.2 Finite Volume Method for Electrolyte Transport

The electrolyte concentration equation is discretized using the **Finite Volume Method (FVM)** across three distinct domains: Anode, Separator, and Cathode. FVM is essential here because mass must be strictly conserved across the material interfaces where porosity (ϵ) and effective diffusivity ($D_{eff} = D\epsilon^b$) change abruptly.

3.2.1 Harmonic Mean for Interface Properties

To ensure continuity of flux across these material interfaces, the **harmonic mean method** is employed for the diffusion coefficients at control volume faces. For a face between node i and $i+1$, the effective diffusivity is calculated as:

$$D^{eff}_{e,i+1/2} = \frac{D^{eff}_{e,i} D^{eff}_{e,i+1}}{\theta D^{eff}_{e,i} + (1-\theta) D^{eff}_{e,i+1}} \tag{12}$$

where θ accounts for the relative distances between nodes and the interface. This approach handles the step-change in transport properties naturally without requiring arbitrary smoothing functions, maintaining high accuracy in the electrolyte concentration gradients which drive the concentration overpotential.

3.3 Matrix State-Space Form

The discretized equations are assembled into a state-space form $\dot{\mathbf{c}} = \mathbf{A}\mathbf{c} + \mathbf{B}u$ to facilitate efficient numerical integration.

For the solid phase diffusion in particle j, the tridiagonal matrix $\mathbf{A}_{s,j}$ represents the diffusion operator:

$$\mathbf{A}_{s,j} = \frac{D_{s,j}}{\Delta r^2} \begin{bmatrix} -2 & 2 & 0 & \dots \\ 1-1/i & -2 & 1+1/i & \dots \\ \vdots & \vdots & \ddots & \vdots \\ 0 & \dots & 2 & -2 \end{bmatrix} \tag{13}$$

The boundary conditions are incorporated as follows:

- **Center ($r = 0$):** The no-flux condition ($\partial c/\partial r = 0$) modifies the first row of $\mathbf{A}$, resulting in coefficients of -2 and 2 due to symmetry.
- **Surface ($r = R_s$):** The flux boundary condition ($\partial c/\partial r \propto I_{app}$) results in a source term that enters the $\mathbf{B}$ vector. The last element of the input vector $\mathbf{B}_{s,j}$ is non-zero:

$$\mathbf{B}_{s,j} = [0, 0, \dots, 2 + \frac{2}{N_r - 1}]^T \tag{14}$$

This vector is multiplied by the input $u = I_{app}$ (scaled by Faraday's constant and geometry) to drive the diffusion dynamics.

3.4 Computational Complexity and State Vector Estimation

The total number of states for a single cell can be estimated by summing the nodes in each discretized domain:

$$N^{cell}_{states} = (N_{r,n} + N_{r,p}) + (N_{x,n} + N_{x,s} + N_{x,p}) + N_{sei} + N_{thermal} + N_{aging} \tag{15}$$

where:

- N_r: Radial nodes in electrode particles (e.g., 100 per electrode)
- N_x: Axial nodes in the electrolyte (e.g., 50 per domain)

- N_{sei}: Nodes for solvent diffusion through SEI (e.g., 5)
- $N_{thermal}$: Thermal states (core/surface temperatures)
- N_{aging}: Scalar states for SEI thickness and capacity loss

For the configuration used in this work ($N_r = 100, N_x = 50, N_{sei} = 5$), a single cell requires approximately 357 states. The total pack state dimension scales linearly with the number of cells: $N_{pack} = N_{cells} \times N_{states}^{cell}$. This linear scaling allows users to accurately predict memory requirements for large pack simulations during the design phase.

4 Pack-Level Modeling Framework

4.1 Electrical Interconnection and Current Distribution

The framework supports arbitrary Series-Parallel configurations.

4.1.1 Kirchhoff's Voltage Law

For a series string of N cells, Kirchhoff's voltage law states:

$$\sum_{k=1}^{N} V_{cell}^{(k)} = V_{pack} \tag{16}$$

4.1.2 Current Distribution in Parallel Cells

For parallel-connected cells within a module, current distributes such that all cells share a common voltage V_{common}. This requires solving a system of nonlinear equations at every time step:

$$V_{cell}^{(k)}(I^{(k)}, \mathbf{x}^{(k)}) - V_{cell}^{(k+1)}(I^{(k+1)}, \mathbf{x}^{(k+1)}) = 0 \tag{17}$$

subject to $\sum I^{(k)} = I_{pack}$.

4.2 Thermal Interconnection Matrix

The framework incorporates a grid-based thermal interconnection model to capture heat transfer between cells. A conductance matrix $\mathbf{G}_{th}$ is constructed based on the pack topology (series and parallel indices).

4.2.1 Conductance Formulation

The thermal conductance between adjacent cells is defined as:

$$G_{base} = \frac{k_{contact} \cdot A_{contact}}{d} \tag{18}$$

where $k_{contact}$ is the thermal conductivity of the interface material, $A_{contact}$ is the contact area, and d is the distance between cell centers.

4.2.2 Heat Flux Calculations

The conductive heat flux $Q_{cond}^{(k)}$ for cell k is calculated by summing the contributions from all neighbors m:

$$Q_{cond}^{(k)} = \sum_{m \in Neighbors} G_{k,m}(T^{(m)} - T^{(k)}) \tag{19}$$

The convective heat flux to the ambient is given by:

$$Q_{conv}^{(k)} = h_{amb} A_{surf} (T^{(k)} - T_{amb}) \tag{20}$$

This matrix formulation allows for the simulation of complex thermal propagation scenarios, where a hotspot in one cell can influence the aging and performance of its neighbors.

4.3 State-Space Formulation

The spatially discretized PDEs are reformulated into a system of ODEs. The state vector for each cell includes solid concentrations, electrolyte concentrations, SEI solvent concentration, SEI thickness, and temperature:

$$\mathbf{x}_{cell} = [\mathbf{c}_s^n, \mathbf{c}_s^p, \mathbf{c}_e, \mathbf{c}_{solv}, L_{sei}, T]^T \tag{21}$$

5 Nonlinear Current Distribution Algorithm

5.1 Problem Formulation

The current distribution problem is formulated as finding the root of a system of nonlinear equations:

$$\mathbf{F}(\mathbf{I}) = \begin{bmatrix} V_{cell}^{(1)}(I^{(1)}) - V_{cell}^{(N_p)}(I^{(N_p)}) \\ \vdots \\ \sum_{k=1}^{N_p} I^{(k)} - I_{total} \end{bmatrix} = \mathbf{0} \tag{22}$$

5.2 Why fsolve is Necessary

The need for a nonlinear solver like `fsolve` arises from the highly nonlinear relationship between current and overpotential in the Butler-Volmer equation:

$$\eta(I) \propto \sinh^{-1}\left(\frac{I}{2ALa_s j_0}\right) \tag{23}$$

Additionally, the open-circuit potential $U(c_s)$ is a nonlinear function of state-of-charge. Because the cell voltage V_{cell} includes these arcsinh terms and OCP nonlinearities, the system of equations equating the branch voltages cannot be solved analytically using linear algebra (e.g., $Ax = b$). Instead, an iterative Newton-based method is required to find the vector of currents $\mathbf{I}$ that satisfies Kirchhoff's laws at each time step.

5.3 Algorithm Flow

The overall simulation algorithm proceeds as follows:

1. **Initialization:**

   - Define pack topology (Series/Parallel configuration).

- Apply Parameter Overrides (e.g., initialize specific cells with different Temperatures or SOCs to simulate heterogeneity).
- Discretize spatial domains (x, r, ξ) and build state-space matrices $(\mathbf{A}, \mathbf{B})$.
- Initialize global state vector $\mathbf{y}_0$ for all cells.

2. **Time Stepping Loop (Integrator):**

   (a) Receive current time t and global state vector $\mathbf{y}$.

   (b) Determine required pack current $I_{pack}(t)$.

   (c) **Solve Current Distribution:** Use `fsolve` to iteratively calculate branch currents $I^{(k)}$ such that all parallel voltages match, accounting for nonlinear overpotentials.

   (d) **Calculate Derivatives:** For each cell k:
   - Compute electrochemical derivatives $(\dot{c}_s, \dot{c}_e)$ using sparse matrix multiplication.
   - Compute aging derivatives $(\dot{L}_{sei}, \dot{c}_{solv})$ based on side reaction current i_s.
   - Compute thermal derivatives $\dot{T}^{(k)}$ using the Thermal Interconnection Matrix.

   (e) Assemble global derivative vector $d\mathbf{y}/dt$.

3. **Output:** Return state history, voltage breakdown, and SEI thickness evolution.

6 Simulation Results and Validation

6.1 Baseline Model Parameters

A representative series-parallel pack is modeled with NMC-graphite chemistry. The simulation code allows for arbitrary pack configurations and parameter overrides, enabling users to define specific heterogeneity patterns. For this study, cell-to-cell variation is introduced through thermal overrides, setting specific cells to higher temperatures (e.g., 310 K) to simulate hotspots.

6.2 Validation Against Physics: PyBaMM Comparison

To validate the core electrochemical physics, the single-cell version of this framework was benchmarked against PyBaMM (Python Battery Mathematical Modelling), an open-source standard for physics-based battery models at 1C Discharge.

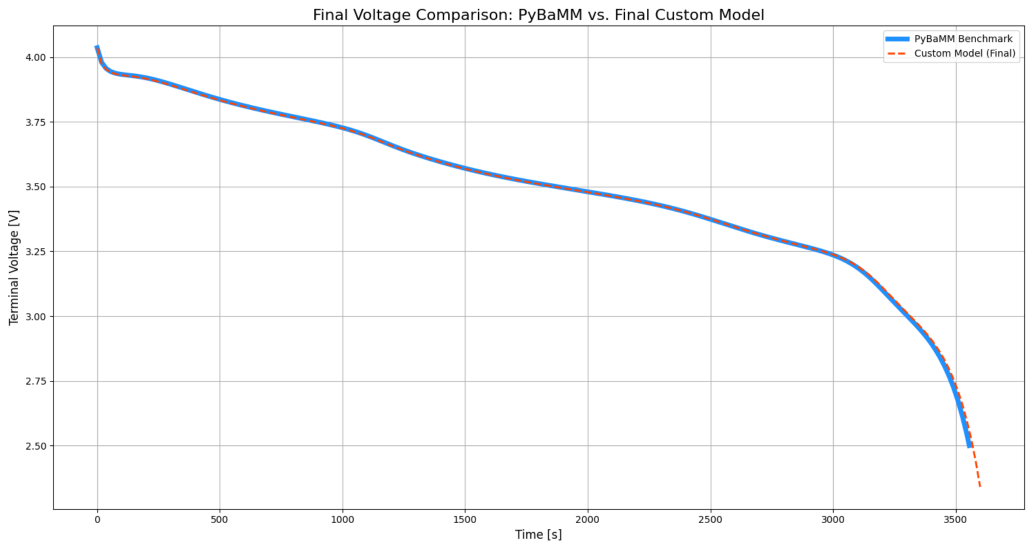


Figure 1: Validation Against PyBaMM
Comparison of Single Cell Voltage Discharge Curve vs. PyBaMM Benchmark

6.3 Single Cell Overpotential Distribution

A detailed breakdown of the cell voltage loss mechanisms was analyzed. The model successfully separates the contributions of Reaction Overpotential, Ohmic Loss (Electrolyte and Solid), Concentration Overpotential, and SEI resistance.

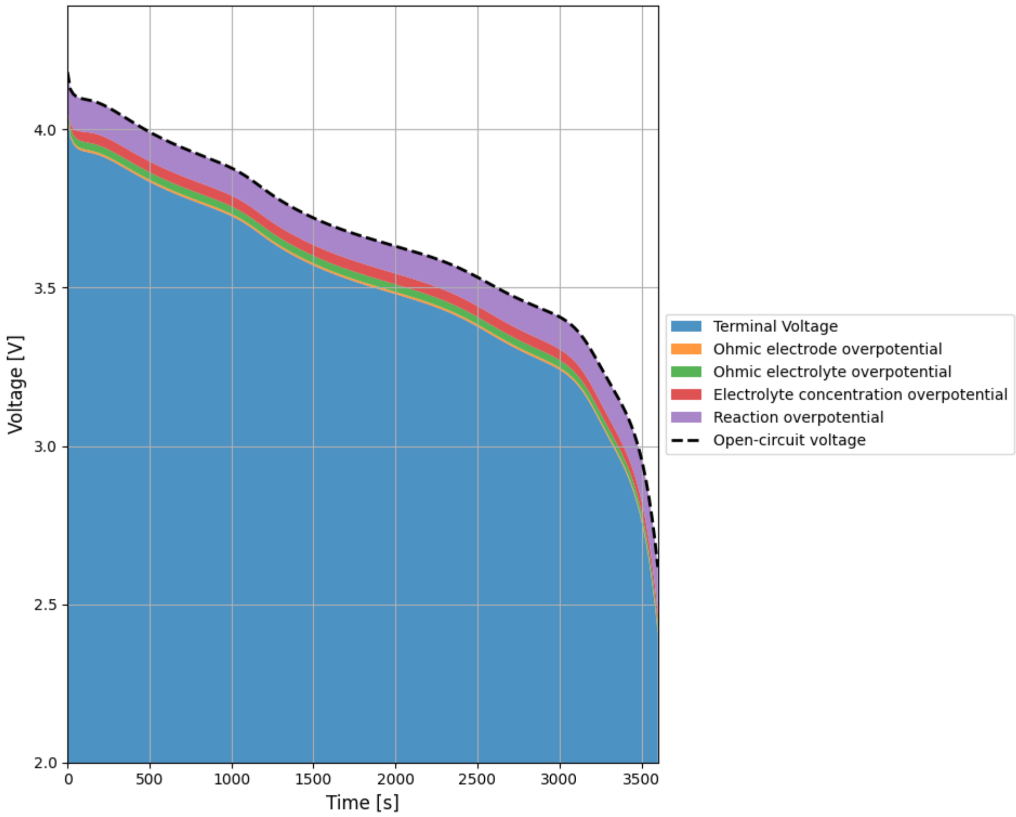


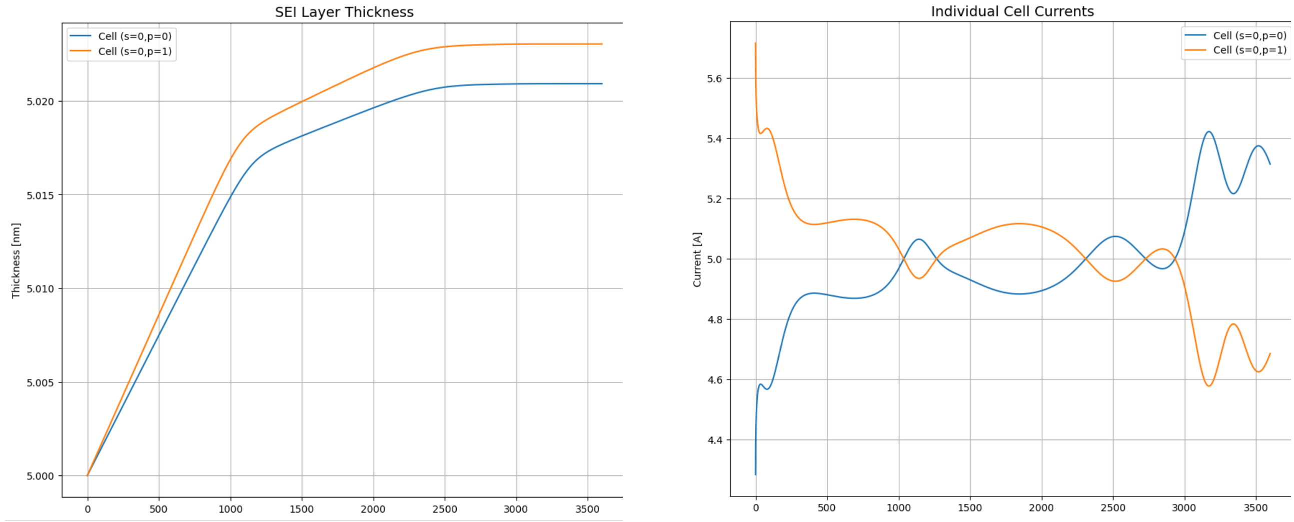
Figure 2: Overpotential Distribution

Stacked plot of voltage loss components (Ohmic, Reaction, Concentration, SEI) over time.

Figure 1: Breakdown of voltage losses within a single cell during discharge. The nonlinear increase in concentration overpotential and reaction overpotential near the end of discharge is clearly captured.

6.4 Effect of Cell Heterogeneity on Degradation

The impact of thermal heterogeneity on aging was investigated by simulating a pack with cells initialized at different temperatures (298.15 K vs 310 K).



(a) SEI Layer Thickness (b) Cell Current

Figure 2: Comparison of Key Pack Diagnostics: (a) Solid Electrolyte Interphase (SEI) Layer Thickness and (b) Cell Current (I_cell) evolution over time for individual cells.

Simulations reveal that hotter cells initially carry higher current due to improved transport kinetics (lower resistance). However, this higher current and temperature accelerate the side reactions, leading to thicker SEI layers. Over long durations, this increased SEI resistance will eventually cause the current to shift away from the aged cells, a complex feedback mechanism successfully captured by this framework.

7 Computational Efficiency

The model is implemented in Python. Currently, a 1-hour simulation for a small pack takes longer than real-time due to the nested solver structure.

- **Bottleneck:** The `fsolve` function is called thousands of times (once per time step) to resolve the nonlinear parallel current distribution.
- **Memory:** State vector size scales linearly with cell count, but remains manageable for mid-sized packs (e.g., ¡100 cells) on standard hardware.

8 Discussion and Future Work

8.1 Key Findings

The developed framework successfully links electrochemical states, thermal conditions, and degradation physics in a pack topology. It highlights that even minor thermal gradients lead to divergent aging paths that cannot be captured by single-cell models.

8.2 Future Work

To address limitations and enable long-term aging studies, future work will focus on:

1. **DAE Formulation:** Reformulating the model from a nested ODE+Algebraic system into a monolithic DAE system to improve speed.

2. **High-Fidelity Python Code:** Developing a robust, open-source Python framework that prioritizes "plug-and-play" modularity. The goal is to allow researchers to easily insert new degradation equations or thermal models without re-writing the core solver, ensuring extensibility.
3. **Computational Efficiency:** Optimizing the solver structure to achieve faster-than-real-time performance, enabling large-scale pack simulations and model-based control applications.
4. **Cooling System Modeling:** Integrating active cooling channel models into the thermal matrix to study thermal management strategies.

9 Conclusion

This report presents a physics-based pack modeling framework that explicitly couples SEI growth, thermal dynamics, and electrical heterogeneity. Validated against PyBaMM, the model provides high-fidelity insights into how thermal hotspots accelerate aging through current redistribution. While currently limited by computational speed, the identified roadmap towards a monolithic DAE formulation and linearized physics lays the groundwork for a versatile, next-generation battery management and research tool.

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A Parameter Summary

Electrochemical Parameters

Parameter	**Symbol**	**Value**	**Unit**
Anode particle radius	$R_{s,n}$	5.86	µm
Cathode particle radius	$R_{s,p}$	5.22	µm
Anode diffusivity	$D_{s,n}$	3.3×10^{-14}	m^2/s
Cathode diffusivity	$D_{s,p}$	4.0×10^{-15}	m^2/s
Max Anode Concentration	$c_{s,max,n}$	33133	mol/m^3
Max Cathode Concentration	$c_{s,max,p}$	63104	mol/m^3
Anode Conductivity	σ_n	215.0	S/m
Cathode Conductivity	σ_p	0.18	S/m
Transference Number	t_+	0.2594	-

Degradation Parameters

Parameter	Symbol	Value	Unit
SEI Kinetics Prefactor	A_{kin}	165.96×10^{-6}	A/m^2
SEI Tunneling Factor	K_{tun}	6.3×10^{9}	m^{-1}
Solvent Diffusivity	D_{solv}	2.5×10^{-22}	m^2/s
SEI Ionic Conductivity	κ_{sei}	5.0×10^{-6}	S/m
SEI Molar Mass	M_{sei}	0.162	kg/mol
SEI Density	ρ_{sei}	1690	kg/m^3